

FOUR PAIRWISE ORTHOGONAL SEMIGROUP OPERATIONS ON A FOUR-ELEMENT SET, AND THE 160 EXTREMAL FAMILIES

ABSTRACT. We settle the single cell $n = 4$ of a problem of Araújo, Bentz, Cameron, Hendrey and Kinyon, who ask, for the set $\mathbb{S}_n$ of semigroup operations on a fixed n -element set, for the maximum size $\max(n)$ of a pairwise orthogonal family and for the extremal families, as functions of n . At $n = 4$ the maximum is 4, and there are exactly 160 labelled extremal families, forming 10 orbits under the diagonal action of $\text{Sym}(4)$, of sizes 8, 8, 8, 8, 8, 24, 24, 24, 24, 24. A family of four operations is exhibited, and the lower bound $\max(4) \geq 4$ is proved by hand: identifying the 4-set with $(\mathbb{F}_2)^2$, the operation $f_M(a, b) = b + M(a)$ is associative exactly when the linear map M is idempotent, and $f_M \perp f_N$ exactly when $M + N$ is invertible. The matching upper bound is a finite exhaustion over all 3492 associative tables on a fixed 4-set; the same census gives $\max(1) = 1$ and $\max(2) = \max(3) = 3$. The classical ceiling for this problem is $N(n) + 2$, which gives only $\max(4) \leq 5$; nothing here settles $n \geq 5$.

1. THE PROBLEM

Araújo, Bentz, Cameron, Hendrey and Kinyon close their paper on complete mappings of semigroups [1] with a list of problems. The seventh of the thirteen problem environments of their closing section of problems reads, character for character (lines 4606–4611 of the L^AT_EX e-print of 2608.25092v1; the sole change is that the map, set inline there, is displayed here):

Problem ([1], seventh problem of the closing list). Let $\mathbb{S}_n$ be the set of semigroup operations on a fixed n -element set. Say that two operations $\cdot$ and $*$ are orthogonal if the map

$$(x, y) \longmapsto (x \cdot y, x * y)$$

from X^2 to X^2 is a bijection. Determine the maximum size of a family of pairwise orthogonal semigroup operations on an n -element set, and identify the extremal families.

The letter X is not bound inside the environment, and we read it as the fixed n -element set. The notation $\mathbb{S}_n$ occurs exactly once in the source, in this Problem, so [1] states no value, no bound, no conjecture and no table for any n . The reading of “orthogonal” as the single joint map $X^2 \rightarrow X^2$ is the

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source's own, matching the cell-by-cell superposition in its worked example of order 3.

Throughout, X is a fixed set with $|X| = n$; $\mathbb{S}_n$ is the set of *all* associative binary operations $X \times X \rightarrow X$, that is all $n \times n$ Cayley tables over X with $(x * y) * z = x * (y * z)$. These are *labelled* operations on a *fixed* set, and no isomorphism quotient is taken anywhere. Write $f \perp g$ for orthogonality. Three immediate consequences of the definition are used below: $\perp$ is symmetric (compose with the coordinate swap of X^2); for $n \geq 2$ no operation is orthogonal to itself, since the image of $(x, y) \mapsto (x \cdot y, x \cdot y)$ lies in the diagonal and has at most $n < n^2$ points; and orthogonality forces both operations to be surjective. Set

$$\max(n) = \max\{|F| : F \subseteq \mathbb{S}_n \text{ pairwise orthogonal}\}.$$

Theorem 1. $\max(4) = 4$. *A pairwise orthogonal family of four semigroup operations on $\{1, 2, 3, 4\}$ is exhibited in Section 2, and no five semigroup operations on a 4-element set are pairwise orthogonal. There are exactly 160 labelled families attaining the maximum; they form 10 orbits under the diagonal action of $\text{Sym}(4)$, of sizes 8, 8, 8, 8, 8, 24, 24, 24, 24, 24, listed in Section 4.*

Proposition 2. $\max(1) = 1$, $\max(2) = \max(3) = 3$, *with respectively 1, 2 and 3 labelled extremal families, in one orbit each. Hence $\max(n) = 1, 3, 3, 4$ for $n = 1, 2, 3, 4$, and both halves of the Problem are answered for every $n \leq 4$.*

2. THE OBJECT

The following four Cayley tables on the fixed set $\{1, 2, 3, 4\}$ are pairwise orthogonal. Rows are indexed by a , columns by b , and the entry is $a * b$.

		1	2	3	4			1	2	3	4
	1	1	1	1	1		1	1	2	3	4
$f_1 =$	2	2	2	2	2		2	1	2	3	4
	3	3	3	3	3		3	3	4	1	2
	4	4	4	4	4		4	3	4	1	2
		1	2	3	4			1	2	3	4
	1	1	2	3	4		1	1	2	3	4
$f_3 =$	2	2	1	4	3		2	4	3	2	1
	3	2	1	4	3		3	1	2	3	4
	4	1	2	3	4		4	4	3	2	1

Concatenating the rows, these are

$$\begin{aligned} &1111222233334444, & 1234123434123412, \\ &1234214321431234, & 1234432112344321, \end{aligned}$$

which is the notation used in the table of Section 4. Here f_1 is the left-zero band $L(x, y) = x$, of Rees–Sushkevich type $(|I|, |G|, |\Lambda|) = (4, 1, 1)$; each of f_2, f_3, f_4 is a right group $\mathbb{Z}_2 \times R_2$, of type $(1, 2, 2)$.

f_4 is published in [1] and is cited, not claimed. At lines 706–720 of the same e-print the authors write: “Using MACE4, we found that the (completely simple) semigroup S given by the Cayley table [1 2 3 4 / 4 3 2 1 / 1 2 3 4 / 4 3 2 1] has eight distinct complete mappings. ... None of those complete mappings preserve $E(S) = \{1, 3\}$.” That is exactly f_4 , and indeed its idempotents are 1 and 3. It is published there for an unrelated purpose and carries no orthogonality claim, so Theorem 1 is untouched; but the table is theirs.

Why this family works, by hand. Identify X with $(\mathbb{F}_2)^2$ by writing 0, 1, 2, 3 for the four elements and $+$ for bitwise XOR, so $a = a_0 + 2a_1$. Then $f_1(a, b) = a$ and $f_i(a, b) = b + M_i(a)$ for $i = 2, 3, 4$, where

$$M_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_4 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

act on $(a_0, a_1)^\top$; concretely M_2 sends 0, 1, 2, 3 to 0, 0, 2, 2, while M_3 sends them to 0, 1, 1, 0 and M_4 to 0, 3, 0, 3. Each M_i is $\mathbb{F}_2$ -linear, of rank 1, and idempotent.

Lemma 3. *Let A be an elementary abelian 2-group and M, N linear endomorphisms of A , and put $f_M(a, b) = b + M(a)$.*

- (i) f_M is associative if and only if $M^2 = M$.
- (ii) $f_M \perp f_N$ if and only if $M + N$ is invertible.
- (iii) $f_M \perp L$ for every M , where $L(a, b) = a$.

Proof. (i) $f_M(f_M(a, b), c) = c + M(b + M(a)) = c + M(b) + M^2(a)$ while $f_M(a, f_M(b, c)) = c + M(b) + M(a)$; these agree for all a, b, c if and only if $M^2 = M$. (ii) The pair map is $(a, b) \mapsto (b + M(a), b + N(a))$. The sum of the two coordinates is $(M + N)(a)$, free of b ; so if $M + N$ is invertible one recovers a from $u + v$ and then b from $u + M(a)$, and conversely if $(M + N)(a) = 0$ with $a \neq 0$ then $N(a) = M(a)$, so the distinct pairs (a, b) and $(0, b + M(a))$ have the same image $(b + M(a), b + M(a))$. (iii) The pair map is $(a, b) \mapsto (a, b + M(a))$, with inverse $(u, v) \mapsto (u, v + M(u))$. $\square$

Part (ii) of Lemma 3 is classical, not ours; see Section 5(C). Part (i) is the restriction to associative operations. Applying the lemma: M_2, M_3, M_4 are idempotent, so f_2, f_3, f_4 are associative; $f_1 = L$ is associative because $(a \cdot b) \cdot c = a = a \cdot (b \cdot c)$ and is orthogonal to each f_{M_i} ; and

$$M_2 + M_3 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad M_2 + M_4 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad M_3 + M_4 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

all have determinant 1 over $\mathbb{F}_2$. So all six pairs are orthogonal and $\max(4) \geq 4$, with no computation involved. The computation of Section 3 supplies only the matching upper bound $\max(4) \leq 4$.

3. THE UPPER BOUND

Lemma 4. *Let $n \geq 2$.*

- (i) (*Balance.*) *If $f \in \mathbb{S}_n$ has an orthogonal partner then every value $a \in X$ occupies exactly n of the n^2 cells of the table of f .*
- (ii) *Two commutative operations are never orthogonal. In particular no two group operations on a 4-set are orthogonal, every group of order 4 being abelian.*
- (iii) *$L(x, y) = x$ and $R(x, y) = y$ are associative with pair map the identity of X^2 , and any group operation g is orthogonal to both; so $\max(n) \geq 3$.*
- (iv) *A family containing both L and R has size at most 3 when $n = 4$.*
- (v) *$\max(n) \leq N(n)+2$, where $N(n)$ is the maximum number of mutually orthogonal Latin squares of order n .*

Proof. (i) The n^2 image pairs are distinct, and those with first coordinate a lie among the n pairs $(a, *)$, so at most n cells carry the value a ; summing n values to n^2 forces exactly n . (ii) (x, y) and (y, x) have the same image. (iii) Rows of a group table are permutations, so $g \perp L$; columns are, so $g \perp R$. (iv) $f \perp L$ and $f \perp R$ together say that every row and every column of f is a permutation, i.e. f is a quasigroup; an associative quasigroup is a group, and by (ii) two group operations are never orthogonal, so at most one further member is possible. (v) Given a pairwise orthogonal family of size $k \geq 2$, re-index the n^2 cells of X^2 by the bijection supplied by two of its members. Those two become the two coordinate projections, and the remaining $k - 2$ operations become $k - 2$ Latin squares, pairwise orthogonal. $\square$

Part (v) is the sharp classical form of the orthogonal-array bound; see Section 5 (A). Since $N(4) = 3$, it gives $\max(4) \leq 5$, and with (iii) the value is pinned only to $3 \leq \max(4) \leq 5$. Section 2 closes the lower half at 4 by hand, so the single statement left for a computation is the negative

no five associative operations on a fixed 4-set are pairwise orthogonal.

The exhaustion, as arithmetic. All 3492 labelled associative operations on the fixed set [4] are generated — no isomorphism quotient, no balance pruning, no lemma used to shrink the vertex set. The count 3492 agrees with the published value $A023814(4)$ [10] of the number of labelled associative binary operations on a 4-set; the completeness of the vertex set, that is the claim that these 3492 tables are all of $\mathbb{S}_4$, is pinned by that agreement rather than proved here, and it cannot be eliminated by any argument inside this note. The orbit count of the diagonal $\text{Sym}(4)$ action on those tables is $188 = A027851(4)$, the number of semigroups of order 4 up to isomorphism, and the orbit sizes sum to 3492, so that decomposition is a partition. Of the 3492 tables, exactly 48 are value-balanced. Each of the remaining 3444 carries some value in at least 5 of its 16 cells, which by Lemma 4(i) leaves it with no orthogonal partner at all: those 3444 vertices have degree 0 and lie in no clique of size ≥ 2 . Restricting to the 48 therefore loses nothing.

Among the 48 there are 354 orthogonal pairs out of all $\binom{48}{2} = 1128$, and the numbers of cliques by size are

$$48, \quad 354, \quad 542, \quad 160, \quad 0.$$

The last two entries are each obtained twice: once by enumerating all cliques level by level, and once by an algorithm-free scan testing every one of the $\binom{48}{4} = 194,580$ four-subsets and every one of the $\binom{48}{5} = 1,712,304$ five-subsets pair by pair. The load-bearing negative therefore does not depend on any clique algorithm. This proves $\max(4) \leq 4$ and, with Section 2, Theorem 1. Running the same census at $n = 1, 2, 3$ gives 1, 8, 113 associative tables, of which 1, 4, 5 are balanced, with 0, 5, 7 edges and maximum cliques of size 1, 3, 3 attained by 1, 2, 3 families; this is Proposition 2.

Remark. The 48 balanced tables are exactly the completely simple ones. This equivalence is nowhere assumed above, and no general form of it is claimed here: it is a checked fact about the census, the accompanying program verifying in both directions, for $n \leq 4$ only, that a table on $[n]$ is value-balanced if and only if $SaS = S$ for every a .

4. ORBIT REPRESENTATIVES AT $n = 4$

At $n = 4$ there are 160 labelled extremal families, in 10 orbits under relabelling of X , of sizes $5 \cdot 24 + 5 \cdot 8 = 160$. Because the Problem asks for families of *labelled* operations, the count is of labelled families and the orbit decomposition is reported in addition to it. Tables are written in the flat row-concatenated form of Section 2; recall $L = 1111222233334444$ and $R = 1234123412341234$.

#	size	contains	representative
1	24	—	1133224411332244, 1212212134344343, 1234123443214321, 1234341234121234
2	24	—	1133224411332244, 1212212134344343, 1234123443214321, 3412123412343412
3	24	—	1133224411332244, 1221122143344334, 1234341234121234, 1414232332324141
4	24	—	1133224411332244, 1221211234434334, 1234123434123412, 1414232332324141
5	24	—	1133224411332244, 1221211234434334, 1234123434123412, 4141323223231414
6	8	L	1111222233334444, 1234123434123412, 1234214321431234, 1234432112344321 (*)
7	8	L	1111222233334444, 1234123434123412, 1234214321431234, 4321123443211234
8	8	a group op	1133224411332244, 1221122143344334, 1234214334124321, 1414323232321414
9	8	R	1133224433114422, 1221211234434334, 1234123412341234, 1414232332324141
10	8	R	1133224433114422, 1221211234434334, 1234123412341234, 4141323223231414

The row marked $(*)$ is the family of Section 2. Of the 160 families, 16 contain L only (orbits 6, 7), 16 contain R only (orbits 9, 10), 128 contain neither, and none contains both, as Lemma 4 (iv) already forces.

5. RELATION TO KNOWN WORK

(A) The ceiling of Lemma 4 (v) is classical in both of its forms and is claimed here in neither. A pairwise orthogonal family of k operations is a set of k columns of an index-1 strength-2 orthogonal array $OA(n^2, k, n, 2)$, whence $k \leq n + 1$ by the Rao and Bose–Bush bounds [11, 6]; the sharper $\max(n) \leq N(n) + 2$ is that bound in Latin-square form, and N is the classical MOLS number, with $N(4) = 3$ and $N(6) = 1$ by Tarry [12]. The source paper’s own introduction works in exactly this orthogonal-mate language and cites Mann [8, 9]. What is claimed here is only the associative determination and the exhaustion.

(B) The phrase “orthogonal systems of operations” names an established literature that [1] does not cite: Belousov [3], Ušan and Stojaković [13], Belyavskaya and Mullen [5], Belyavskaya [4], Fryz [7]. That school studies orthogonality of k -invertible, quasigroup and n -ary operations, and proves criteria and coordinatisation theorems; the framework terminology is theirs. None of it imposes associativity, and none of the items retrieved states a maximum at order 4. Of them, Fryz [7] stands closest to an extremality question, its subject being the number of ways an orthogonal set of operations can be completed, but the operations completed there are k -invertible rather than associative. Adjacent from the other side, and restricted to Cayley tables of groups, are Bailey, Cameron, Kinyon and Praeger [2] and Evans, Martin, Minden and Ollis (arXiv:1807.08580); neither addresses $\mathbb{S}_n$, nor either half of the Problem.

(C) Lemma 3 (ii) belongs to that school. It is the case $A = M$, $B = I$, $C = N$, $D = I$ of the classical criterion for orthogonality of *linear* operations $f(x, y) = Ax + By$ over an abelian group, and it is stated here as a special case of a known criterion rather than as a new observation. We do not attribute it to a particular statement in the literature of (B). Part (i) of the same lemma is the associativity condition $M^2 = M$, which is what confines the family used here to $\mathbb{S}_4$.

6. WHAT IS NOT SETTLED

The Problem asks for $\max(n)$ and the extremal families as functions of n . What is settled here is the cells $n \leq 4$: $\max(n) = 1, 3, 3, 4$ with 1, 2, 3, 160 extremal families in 1, 1, 1, 10 relabelling orbits. Nothing above closes $n \geq 5$, and the result should not be quoted as closing the Problem: no table of order above 4 is generated anywhere by the accompanying program, and no value of $\max(n)$ for $n \geq 5$ is claimed. The ceiling of Lemma 4 (v) is quoted from the classical literature and is not reproved here.

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